



People's Education Trust(R.),
P.E.S. College of Engineering, Mandya

(An Autonomous Institution Aided by Government of Karnataka, Affiliated to VTU, Belagavi)

NEXO VATE ROBO SUMO



OBJECTIVES:

Design and build a robot (manual or autonomous) that can push the opponent's robot out of a circular arena while following all rules and constraints.

ELIGIBILITY:

- The competition is open to all students.
- Each team must consist of 2 to 4 members.
- A participant cannot be a part of multiple teams.

Open to all students from any college or institution. Participants can compete individually or as a team, as per event rules. All participants must carry a valid student ID card for verification. Entry will be permitted only after successful registration. Participants are expected to follow all event guidelines and maintain discipline throughout the competition.



22ND & 23RD

MAY 2026



VENUE

HDC AUDITORIUM

PES COLLEGE OF ENGINEERING



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NEXOULATE

BOT SPECIFICATIONS:

- Control Type: Wired or wireless.
- Maximum Weight: 5 kg (with 5% tolerance, up to 5.25 kg).
- Maximum Dimensions: 30 × 30 × 30 cm (with 5% tolerance, up to 31.5 cm).
- Construction: Robots must be fully custom-built. LEGO, kits, or ready-made bots are not allowed.
- Power Source: Only electrical power is allowed (no IC engines).
- Power Supply: Onboard battery or wired external source permitted.
- Voltage Limit: Maximum 12.5V across any two points.
- Backup: Teams must carry spare batteries.
- Wired Bots: Cable must be long enough and remain slack during the match.

MATCH FORMAT:

- Each match consists of 2 rounds (1.5 minutes each).
- In case of a tie, an additional 1-minute round will be played.
- Winning a round earns 1 point.
- The robot that pushes the opponent completely out wins the round.
- A robot is considered out only when it fully falls off the platform.
- If both robots fall, the one touching last wins.
- If a robot becomes immobile, the opponent wins the point.
- If no result, a 2-minute sudden-death round decides the winner.



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RESTART & TIMEOUT RULES:

- If robots are stuck for more than 10 seconds, the match will restart.
- Each team is allowed one 60-second technical timeout per match.

RULES & FAIR PLAY:

- Teams must have 2–4 members.
- Any misconduct or intentional damage leads to disqualification.
- Interference with opponent robots is strictly prohibited.
- Judges' decisions are final and binding.
- Organizers may modify rules if required.
- Matches are controlled only by referee instructions.



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